Joseph Havens

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EDUCATION

Bachelor of Science in Physics and Astronomy, Minor in German
Expected Graduation: May 2027
University of Kansas

RESEARCH EXPERIENCE

Undergraduate Research Assistant December 2024-Present
University of Kansas Advisors: Dr. Allison Kirkpatrick, Dr. Bren Backhaus

- **Project: Dust Attenuation in SMACS and JADES Galaxies**
  - **Goal:** Examining how dust attenuation affects a galaxy's luminosity in the distant universe (>7 billion years ago) using observations from the James Webb Space Telescope.
  - Analyzing and classifying spectra from GLASS and JADES (two extragalactic fields observed with JWST) using multi-line attenuation modeling.
  - **Notable Achievement:** Formulated a two-tier, rigorous methodology for identifying Active Galactic Nuclei (AGN) and AGN-Candidates around $z \sim 2$ utilizing machine learning, multi-wavelength observations, and spectral analysis.
- **Project: Zooniverse Site Development** Advisors: Dr. Allison Kirkpatrick, Greg Troiani
  - **Goal:** Contributing to and refining the research team's Zooniverse site (`www.zooniverse.org` - Cosmic Collisions).
  - Use TRILOGY (proprietary image conversion software) to prepare .fits files for submission to the Zooniverse site.
  - **Result:** Created multi-color cutouts of distant galaxies, and the Zooniverse project resulted in the classification of 10,000 galaxies.

Undergraduate Research Assistant February 2023-August 2024
University of Kansas Advisor: Dr. Steven Prohira

- **Goal:** Built, ran, and analyzed simulations of radio wave propagation through a medium of ice with a changing index of refraction to establish a model for later comparison.
- Use *ParaPropPython* (proprietary Parabolic Equation simulation), MEEP (FDTD computational electromagnetic simulation), and ray tracing to compare and analyze various simulation methods for radio wave propagation.

PUBLICATIONS

Co-Led

- **Havens, J.**, Zafuta, O., & Backhaus, B. E., 2026, *Constraining Dust Attenuation Curves in Star Forming Galaxies Around Cosmic Noon Using Multiple Emission Lines*, in prep.

Significant Author

- Troiani, G., Kirkpatrick, A., **Havens, J.**, & Bajaj, L., 2026, *Cosmic Collisions: Visual and Non-parametric Optical Morphologies for 14000 Galaxies from PRIMER*, in prep.

- Siwakoti, U., Wright, S., Shaw, N., Zaman, M. A. A., Lieber, A. E., Sheriff, K. E., Mai, X., **Havens, J.**, Mills, E., Cionitti, R., & Rhem, M., 2026, *A JWST MIRI-MRS Map of the Nucleus of the Nearby Starburst Galaxy NGC 253*, in prep.
- Sheriff, K. E., Mai, X., Lieber, A. E., Smith, K., Siwakoti, U., Mills, E. A. C., Wright, S., Rhem, M., **Havens, J.**, & Zaman, M. A. A., 2026, *A Young Superbubble in NGC 253's Central Starburst*, in prep.
- Bajaj, L., Troiani, G., Stephens, J., Kirkpatrick, A., Cacho-Negrete, C., Cionitti, R., Shaw, N., **Havens, J.**, and other authors, 2026, *JWST Discovers a Candidate Obscured AGN Offset 3.5kpc from its center at $z \sim 0.58$*, in prep.

PRESENTATIONS

- **Why Galaxies Lie to Us**, Presented to PHSX 150 (Freshman Physics and Astronomy Seminar), University of Kansas — Nov. 20, 2025
- **Why Galaxies Lie to Us** (updated), Presented to Society of Physics Students (SPS), University of Kansas — Mar. 4, 2026

TEACHING & OUTREACH

Undergraduate Teaching Assistant (UGTA) — Fall 2025
University of Kansas

- ASTR 591 - Stellar Astronomy - Dr. Allison Kirkpatrick
- Graded problem sets for 40+ students.
- Advised students through computational astronomy assignments.

Undergraduate Teaching Assistant (UGTA) — Spring 2026
University of Kansas

- PHSX 313 - Modern Physics - K.C. Kong
- Graded quizzes and provided assistance as needed outside of class.

Astronomy Merit Badge Instructor — Feb. 7, 2026
Scouts BSA

- Developed and led a curriculum to teach foundational astronomy concepts and observational skills.
- Mentored three middle-school students in foundational astronomy, successfully fostering a deep interest in astrophysics (particularly black hole dynamics) and inspiring a self-reported desire to pursue future Astronomy careers.

Public Night Volunteer — Spring 2025-Present
KU Physics & Astronomy Department

- Operated telescopes and guided public observations during community outreach events.

Sunset Hill Elementary — Apr. 10, 2026
Sunset Hill Elementary School, Lawrence, KS

- Demonstrated and encouraged a fascination of physics to elementary students through the use of a Van de Graaff Generator.

Ad Astra Kansas Apr. 11, 2026
Washburn University, Topeka, KS

- Celebrated 250 years of STEM and Science, hosted by Washburn University.
- Engaged the public in demonstrations of magnetism and light.
- Represented the department in local media interviews regarding the importance of STEM education and the next generation of scientists.

Hillcrest Elementary Apr. 15, 2026
Hillcrest Elementary School, Lawrence, KS

- Educated elementary students on the principles of electrostatics through live demonstrations using a Van de Graaff generator.
- Designed engaging, hands-on learning experiences to bridge the gap between abstract physics concepts and real-world applications.

RESEARCH IN COURSEWORK

(Graduate) Radiation and Interstellar Medium-ASTR 796 Fall 2025

- **Project:** Class project with multiple groups including Polycyclic Aromatic Hydrocarbons (PAH) / silicate lines in the nearby galaxy NGC 253.
  Title: A JWST MIRI-MRS Map of the Nucleus of the Nearby Starburst Galaxy NGC 253 by E.A.C. Mills et al
- **Skills:** Graduate-level education in atomic transitions, Polycyclic Aromatic Hydrocarbons (PAHs), emission lines (specifically $7.7 \mu m$ and $11 \mu m$ PAH lines), and Integral Field Units (IFUs - instruments that capture spatial and spectral data simultaneously).

Stellar Astronomy-ASTR 591 Fall 2023

- **Project:** Analyzed Hertzprung-Russell (HR) diagrams of stellar clusters to determine age.
- **Result:** Found significant agreement with literature consensus, with ages around $\log_{10}(\text{age}) \approx 8-10$.
- **Skills:** Improved Python skills, engaged with Astronomy literature, developed scientific writing skills.

Observational Astrophysics-ASTR 596 Fall 2023

- **Project:** Use the Breyo Observatory at Sienna College in Loudonville, NY for remote sky imaging.
- **Result:** Extracted and analyzed astronomical spectra from various targets, utilizing Python to perform robust emission and absorption line (EL/AL) fitting.
- **Skills:** Developed skills in Jupyter Notebook, GitHub, bash commands, and remote telescope operation.

SKILLS & ABILITIES

- **Languages:** English (Native), German (Professional Working Proficiency), French (Professional Reading Proficiency), Spanish (Elementary)
- **Programming Languages:** Python (proficient), LaTeX (proficient), Arduino variant of C++ (intermediate), bash/Zsh (beginner)
- **Data Science Libraries:** NumPy, Pandas, Matplotlib, SciPy, Scikit-learn

- **Astro-Specific Libraries:** Astropy, Specutils, AGNBoost, LMFIT, CLOUDY
- **Computation & Imaging:** Numba (JIT performance optimization), OpenCV (image processing), Git/Github
- **Cloud Data & APIs:** Google Cloud Platform (GCP), Google Sheets API (gspread), building data pipelines with REST APIs (requests, json)
- **Machine Learning:** Frameworks (TensorFlow); Models (Artificial Neural Networks)

LEADERSHIP & ACTIVITIES

Mailroom Attendant — August 2025 - Present
Department of Physics and Astronomy

Attendee, PHYSCon 2025 — Oct. 29 - Nov. 1, 2025
Denver, CO.

Society of Physics Students (SPS), KU Chapter — Fall 2024-Present

- Member, Co-led logistics for conferences.

Scouting America - Troop 249, Jayhawk Area Council — Feb. 2014 - June 30, 2020

- Attended and later staffed the National Youth Leadership Training (NYLT), developing skills in leadership, presentation, cooperation, and quick decision making.
- Led a troop of 30+ scouts for three consecutive 6-month terms.
- Served as Junior Assistant Scout Master (JASM), advising youth leadership and assisting adult leadership.

University of Kansas Men's Volleyball Club — Fall 2024

CERTIFICATIONS AND AWARDS

- Eagle Scout - June 2020, 5 Eagle Scout Palms